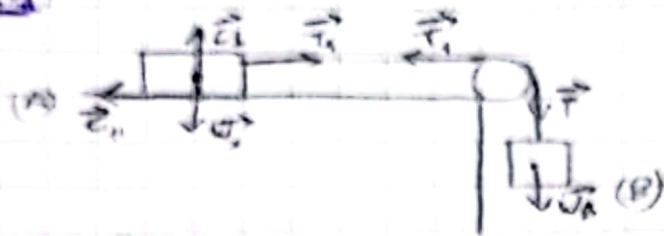


Exercise Part II

Ex1:



1) acceleration:

object (A): $\sum \vec{F} = m\vec{a}$
 $\vec{C}_1 + \vec{C}_2 + \vec{W}_A + \vec{T} = m\vec{a}$
 $T - C_1 = m_A a \quad (1) \quad \left(\frac{W_A}{W_B} = \frac{C_1}{C_2} \right)$

object (B): $\vec{W}_B + \vec{T} = m_B \vec{a}$
 $W_B - T = m_B a \quad (2)$

(1) + (2): $W_B - C_1 = (m_A + m_B) a$
 $a = \frac{W_B - C_1}{(m_A + m_B)} = 0,2 \text{ (m/s}^2\text{)}$

$\Rightarrow C_1 = \mu_k \cdot C_2 = \mu_k \cdot m_A g = 3,924 \text{ (N)}$

2) Tension:

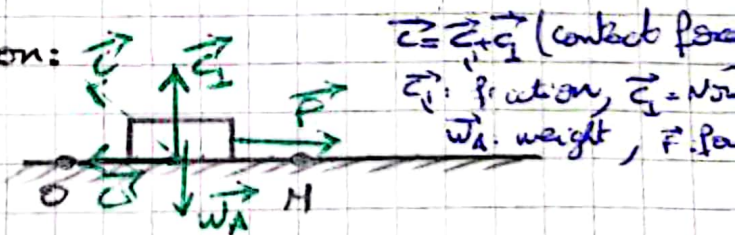
$T = m_A a + C_1 = 4,32 \text{ (N)}$

3) distance: $x = \frac{1}{2} a t^2 + v_0 t + x_0$
 $= \frac{1}{2} (0,2) (2)^2 = 0,4 \text{ m}$

Ex2:

* minimum force to start the motion:

$f_m = C_{s1} = \mu_s \cdot C_{10} = \mu_s \cdot m g$
 $= 0,6 \times 2 \times 10 = 12 \text{ N}$



1) acceleration:

$\sum \vec{F} = m\vec{a} \Rightarrow \vec{C}_1 + \vec{C}_2 + \vec{F} + \vec{W}_A = m\vec{a}$
 $\vec{C}_1 + \vec{C}_2 + \vec{F} + \vec{W}_A = m\vec{a}$

$-C_1 + F = m_A a \quad (\vec{C}_1 = -\vec{W}_A)$
 $\Rightarrow a = \frac{F - C_1}{m_A}$

$C_1 = \mu_k \cdot C_2 = \mu_k \cdot m_A g = 8 \text{ N}$

$a = \frac{14,8}{2} = 7,4 \text{ m/s}^2 \text{ (Non uniform motion)}$

$x = \frac{1}{2} a t^2 + v_0 t + x_0 = \frac{3}{2} t^2 + v_0 t$

2) velocity U_m :

$$U_m^2 - U_0^2 = 2ad \Rightarrow U_m = \sqrt{2ad + U_0^2} = 5.84 \text{ (m/s)}$$

3) velocity U_{A+B} :

$$P_{\text{before}} = P_{\text{after}}$$

$$m_A \vec{U}_A + m_B \vec{U}_B = (m_A + m_B) \vec{U}_{A+B} \Rightarrow \vec{U}_{A+B} = \frac{m_A \vec{U}_A}{(m_A + m_B)} = 1.168 \text{ m/s}$$

$$(U_{A+B} = 1.168 \text{ m/s})$$

Ex 3:

$$\vec{C}_L = -\vec{W}_A$$

$$C_L = W_A$$

$$\frac{C_{11}}{n_A}$$

$$= \mu_k m_A g = 8 \text{ N}$$

$$= 3 \text{ m/s}^2$$

or motion)

$$+ v_0 t + a_0$$

$$v_0 t$$

$$e$$

$$m$$

$$ad$$

$$2ad + v_0^2$$

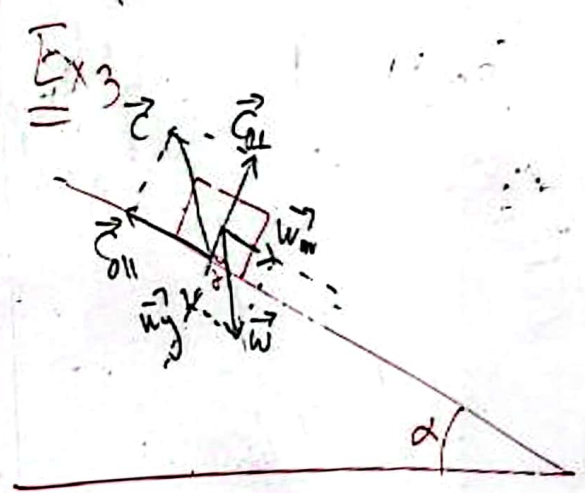
$$84 \text{ (m/s)}$$

$$-v_{A+B}$$

$$m_A \vec{v}_A + m_B \vec{v}_B = (m_A + m_B) \vec{v}_{A+B}$$

$$\vec{v}_{A+B} = \frac{m_A \vec{v}_A}{(m_A + m_B)} = 1,168 \text{ m/s}$$

$$(v_{A+B} = 1,168 \text{ m/s})$$



1) $\sum \vec{F} = \vec{0}$

$$\vec{C} + \vec{W} = \vec{0}$$

$$\vec{C}_{11} + \vec{C}_{12} + \vec{W} = \vec{0} \quad (\vec{W} = \vec{W}_x + \vec{W}_y)$$

$$C_{11} = \mu_s C_{12} = \mu_s \cdot mg \cos \alpha$$

$$C_{01} = W_y = mg \cos \alpha$$

$$W_x = mg \sin \alpha$$

$$\left(\begin{matrix} C_{01} = W_y \\ C_{02} = -W_y \end{matrix} \right)$$

$$-C_{01} + C_{02} + W_x - W_y = 0$$

$$W_x - C_{01} = 0$$

$$mg \sin \alpha - \mu_s mg \cos \alpha = 0$$

$$\sin \alpha = \mu_s \cos \alpha$$

$$\mu_s = \frac{\sin \alpha}{\cos \alpha} = \tan \alpha$$

$$\alpha = 21.8^\circ$$

2) Acceleration

$$\sum \vec{F} = m \vec{a}$$

$$\vec{C} + \vec{W} = m \vec{a}$$

$$\Rightarrow -C_{11} + C_{12} + W_x - W_y = ma$$

$$C_{11} = \mu_k \cdot mg \cos \alpha$$

$$\Rightarrow mg \sin \alpha - \mu_k mg \cos \alpha = ma$$

$$a = g \sin \alpha - \mu_k g \cos \alpha$$

$$= -0,3 \text{ m/s}^2$$

distance:

$$v_f^2 - v_i^2 = 2ad$$

$$\Rightarrow d = \frac{v_f^2 - v_i^2}{2a} = 0,66 \text{ m}$$

2) For $v = \text{crit}$

$$\sum \vec{F} = m \vec{a} = \vec{0} \quad v = \text{crit}$$

$$\vec{C} + \vec{W} + \vec{F} = \vec{0}$$

$$mg \sin \alpha - \mu_k mg \cos \alpha - F = 0$$

$$F = mg \sin \alpha - \mu_k mg \cos \alpha$$

$$= m a = -0,3 \text{ N}$$

Dynamics (Part II.)

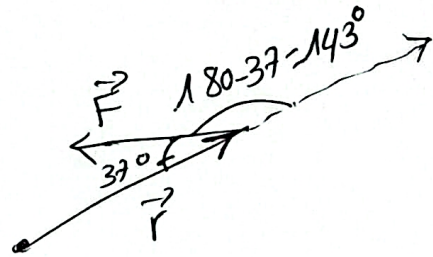
E_{x4}

1) Torque:

$$\vec{\tau} = \vec{r} \wedge \vec{F}$$

$$\tau = r \cdot F \cdot \sin \theta$$

$$= 0,25 \times 17 \times \sin 143^\circ = 2,55 \text{ (N.m)}$$



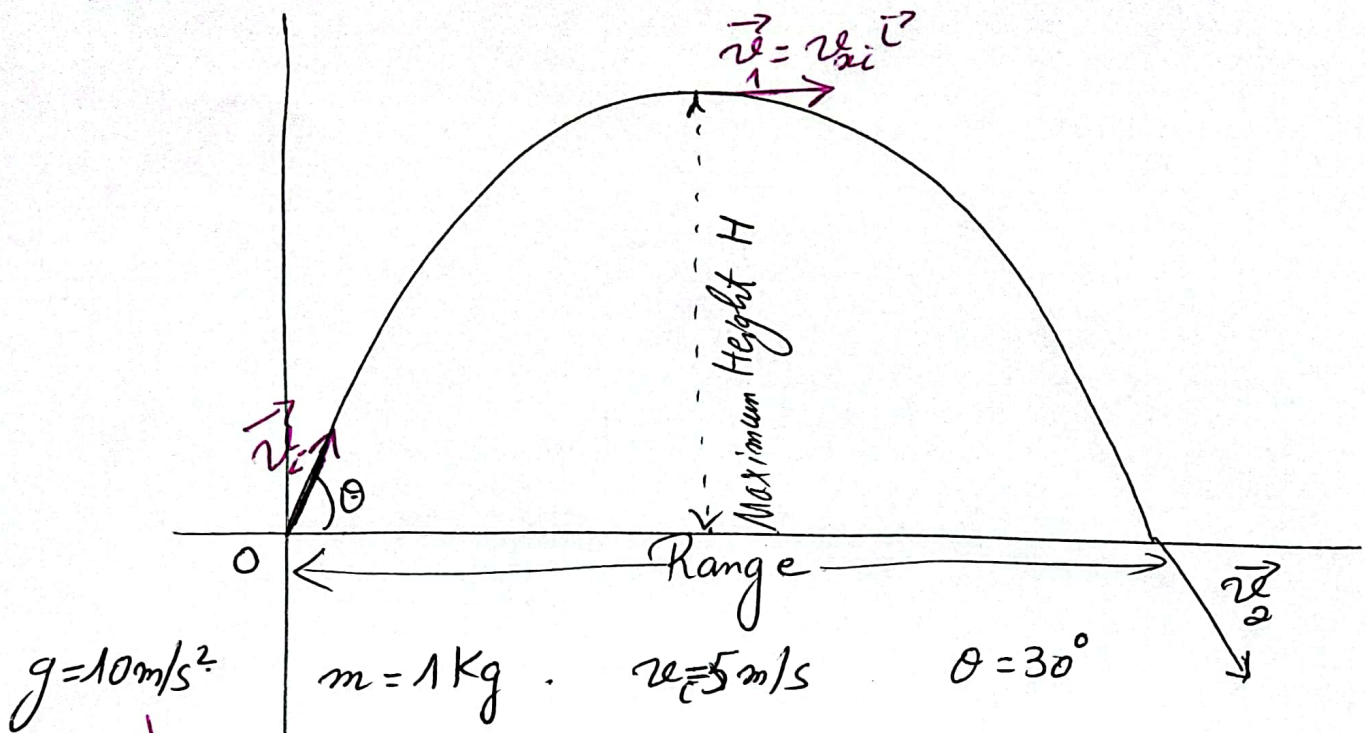
2) Maximum Torque:

$$\sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \tau_{\max} = r \cdot F \cdot \sin \frac{\pi}{2} = 0,25 \times 17 \times 1 = 4,25 \text{ (N.m)}$$

the force should be $\perp$ to the wrench.

Ex 5



1) Angular momentum:

$$\vec{L} = \vec{r} \wedge \vec{p}$$

$\vec{L}$: Angular momentum.

$\vec{r}$: position vector

$\vec{p}$: Linear momentum

$$\vec{r} = x \vec{i} + y \vec{j}$$

$$\vec{p} = p_x \vec{i} + p_y \vec{j}$$

$$\vec{L} = \vec{r} \wedge \vec{p} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & 0 \\ p_x & p_y & 0 \end{vmatrix} = (x p_y - y p_x) \vec{k}$$

$$L = x p_y - y p_x$$

January 2025 (NHSM)

Page 2

Ex 5

$$L = x p_y - y p_x$$

$$p_y = m v_y$$

$$p_x = m v_x$$

$$L = m(x v_y - y v_x)$$

$$x = v_i \cos \theta t \quad \dots (1)$$

$$y = -\frac{1}{2} g t^2 + v_i \sin \theta t \quad \dots (2)$$

$$v_x = \frac{dx}{dt} = v_i \cos \theta \quad \dots (3)$$

$$v_y = \frac{dy}{dt} = -g t + v_i \sin \theta \quad \dots (4)$$

projectile motion equations.

a) At the origin: $L = 0 \text{ (kg} \cdot \frac{\text{m}^2}{\text{s}})$ because $x = y = 0 \text{ (m)}$.

b) At the Highest point:

At the highest point the velocity $v_y = 0 \text{ m/s}$

$$\text{so } L = -m y v_x$$

We have $v_x = v_i \cos \theta$.

$y = H$ (Maximum Height) .. to calculate its value we must determine the time to reach this point using the velocity formula:

$$v_y = 0 = -g t_{\frac{1}{2}} + v_i \sin \theta \Rightarrow t_{\frac{1}{2}} = \frac{v_i \sin \theta}{g} \quad \dots (5)$$

$$\text{⑤ in eq (2)} \Rightarrow y = H = \frac{v_i^2 \sin^2 \theta}{2g}$$

$$\Rightarrow L = -m y v_x = -m \left(\frac{v_i^2 \sin^2 \theta}{2g} \right) (v_i \cos \theta) = -\frac{m v_i^3 \sin^2 \theta \cos \theta}{2g}$$

$$\Rightarrow L = \frac{-m v_i^3 \sin^2 \theta \cos \theta}{2g} = -1.34 \text{ (kg} \frac{\text{m}^2}{\text{s}})$$

Ex

c) before it hits the ground:

In this case $y = 0$ (m).

and the angular momentum formula is:

$$|L = m r v_y|$$

for calculate L we should determine r and v_y

for this we use the time of flight value:

$$\text{we have } t_f = 2 t_{\frac{1}{2}} = \frac{2 v_i \sin \theta}{g} \quad \text{--- (6)}$$

$$\text{(6) in Eq (1)} : r = v_i \cos \theta \left(\frac{2 v_i \sin \theta}{g} \right) = \frac{v_i^2 \sin 2\theta}{g} \quad (\text{r is the range})$$

$$\text{(6) in eq (4)} \quad v_y = v_{y2} = -g \left(\frac{2 v_i \sin \theta}{g} \right) + v_i \sin \theta = -v_i \sin \theta$$

$$\Rightarrow L = m r v_y = -m \frac{v_i^3 \sin 2\theta \sin \theta}{g} = -5,37 \text{ (Kg} \cdot \frac{\text{m}^2}{\text{s}})$$

$$L = - \frac{m v_i^3 \sin 2\theta \sin \theta}{g} = -5,37 \text{ (Kg} \cdot \frac{\text{m}^2}{\text{s}})$$

3) Torque τ :

$$\tau = \frac{dL}{dt}$$

$$L = m(x v_y - y v_x)$$

$$\begin{aligned} L &= m \left[(v_i \cos \theta t)(-gt + v_i \sin \theta) - \left(-\frac{1}{2} g t^2 + v_i \sin \theta t \right) (v_i \cos \theta) \right] \\ &= m \left[-g v_i \cos \theta t^2 + v_i^2 \cos \theta \sin \theta t + \frac{1}{2} g t^2 v_i \cos \theta - v_i^2 \cos \theta \sin \theta t \right] \\ &= -\frac{1}{2} m g v_i \cos \theta t^2 \end{aligned}$$

$$\Rightarrow \tau = \frac{dL}{dt} = \frac{d \left(-\frac{1}{2} m g v_i \cos \theta t^2 \right)}{dt} = -m g v_i \cos \theta$$

↑ this causes angular momentum to change

January 2021 (NHSM)

Princip

Ex 6

a) magnitude of the angular momentum L

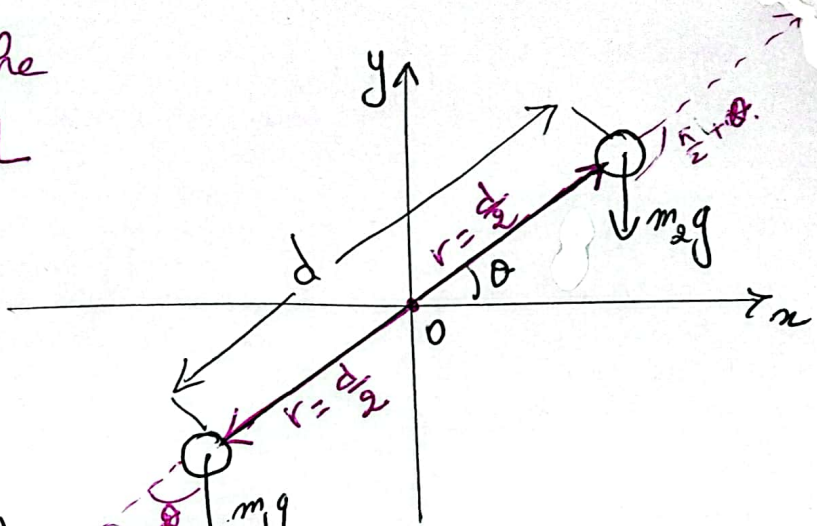
$$L_{\text{tot}} = I_{\text{tot}} \omega$$

$$= (I_{m_1} + I_{m_2} + I_{\text{rod}}) \omega$$

$$= \left(m_1 r^2 + m_2 r^2 + \frac{M d^2}{12} \right) \omega$$

$$= \left(m_1 \left(\frac{d}{2} \right)^2 + m_2 \left(\frac{d}{2} \right)^2 + \frac{M d^2}{12} \right) \omega = \frac{d^2}{4} \left(m_1 + m_2 + \frac{M}{3} \right) \omega$$

$$\boxed{L_{\text{tot}} = \frac{d^2}{4} \left(m_1 + m_2 + \frac{M}{3} \right) \omega}$$



b) Angular acceleration $\ddot{\theta}$ ($\ddot{\theta} = \alpha$).

$$\text{We have } L = I\omega \Rightarrow \frac{dL}{dt} = I \frac{d\omega}{dt} \Rightarrow \tau = I \ddot{\theta}$$

$$\Rightarrow \ddot{\theta} = \frac{\tau}{I} \Rightarrow \ddot{\theta} = \alpha = \frac{\tau_{\text{tot}}}{I_{\text{tot}}}$$

$$\tau_{\text{tot}} = \tau_{m_1} + \tau_{m_2} + \tau_{\text{rod}} \quad (\text{Torque})$$

$$\tau_{m_1} = \frac{d}{2} m_1 g \sin \left(\frac{\pi}{2} - \theta \right) = \frac{d}{2} m_1 g \cos \theta$$

$$\tau_{m_2} = -\frac{d}{2} m_2 g \sin \left(\frac{\pi}{2} + \theta \right) = -\frac{d}{2} m_2 g \cos \theta$$

$$\Rightarrow \tau_{\text{tot}} = \frac{d}{2} \cos \theta g (m_1 - m_2)$$

$$\Rightarrow \ddot{\theta} = \frac{\frac{d}{2} \cos \theta g (m_1 - m_2)}{\frac{d^2}{4} \left(m_1 + m_2 + \frac{M}{3} \right)} = \frac{2 \cos \theta g (m_1 - m_2)}{d \left(m_1 + m_2 + \frac{M}{3} \right)}$$

January 2025 (NHSM)

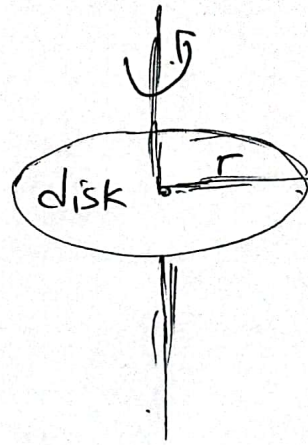
Page 5

Ex7

$$m = 3 \text{ Kg}$$

$$\omega = 6 \text{ rad/s}$$

$$r = 0,2 \text{ m}$$



1) Angular momentum:

a) Axis of rotation passes through its center of mass:

$$L = I_0 \cdot \omega = \frac{mr^2}{2} \cdot \omega = 3 \times \left(\frac{0,2}{2}\right)^2 \times 6 = 0,36 (\text{Kg} \cdot \frac{\text{m}^2}{\text{s}})$$

where: L is angular momentum

ω is angular velocity

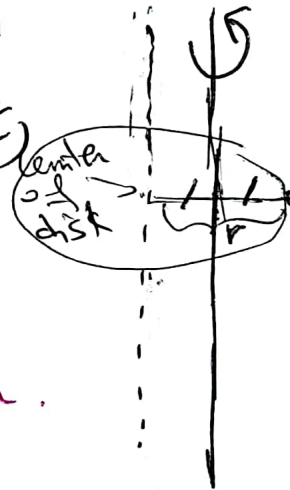
I_0 moment of inertia (At center).

b) Axis of rotation passes through a point midway between the center and the rim

$$L = I \omega = \left(\frac{3mr^2}{4}\right) \omega = 0,54 (\text{Kg} \cdot \frac{\text{m}^2}{\text{s}})$$

$$I = I_0 + m\left(\frac{r}{2}\right)^2 = \frac{mr^2}{2} + \frac{mr^2}{4} = \frac{3mr^2}{4}$$

Theorem of Hygens-Steiner.



January 2025 (NHSM)

Page 6